



MIMIR - Can We Vibe-Code Hardware Too?

Giacomo Brunetta, Adarsha Balaji, Christopher Knight, Zhiling Lan, Valerie Taylor
University of Illinois Chicago, Argonne National Laboratory

Abstract

MIMIR is an **automated hardware generation framework** that **integrates High-Level Synthesis (HLS)** with **multi-agent generative AI** to produce **FPGA accelerators from high-level software descriptions**. By combining compiler-driven optimizations with agent-based design space exploration, MIMIR targets high Quality of Results (QoR) while removing the need for specialized hardware design expertise.

Motivation

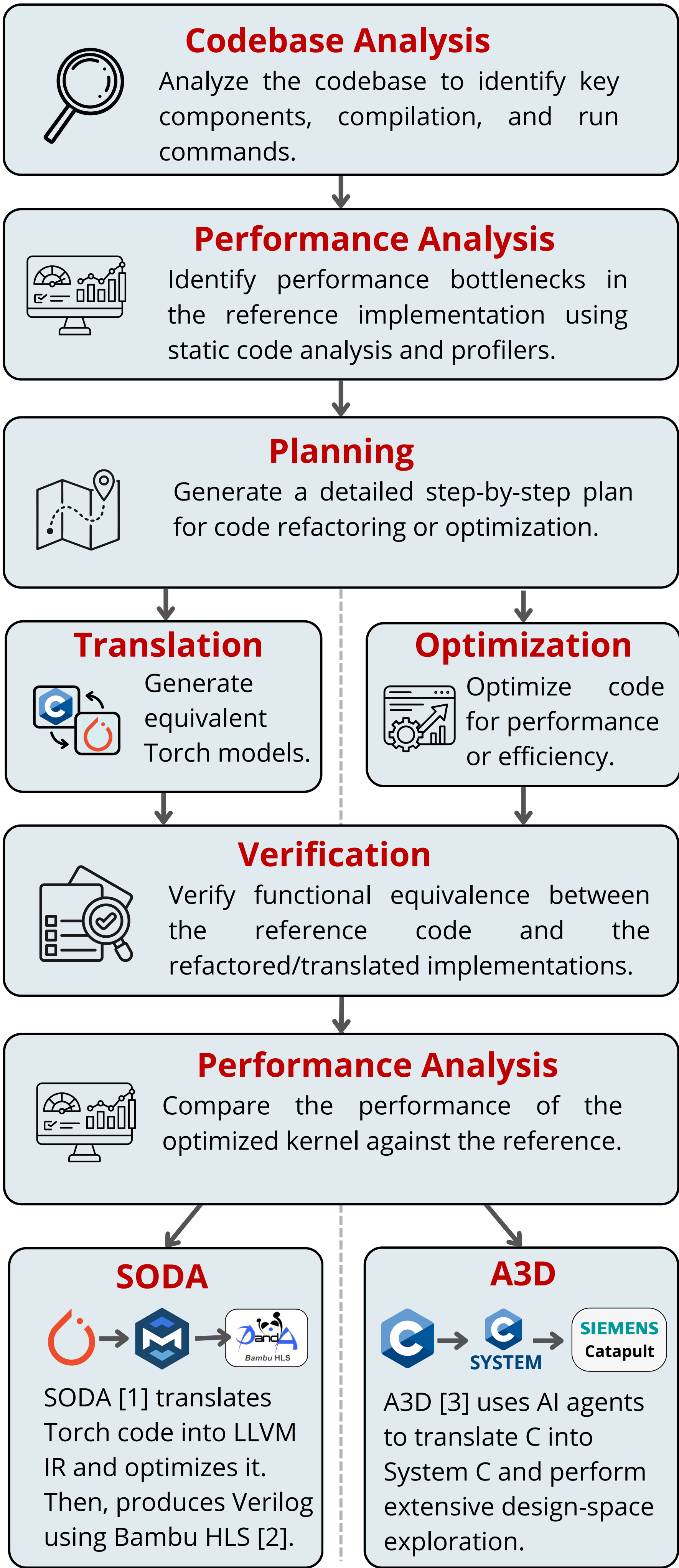
The end of Moore’s Law scaling has shifted performance gains toward domain-specific architectures. Custom hardware accelerators offer significant improvements in throughput and energy efficiency, but their design remains costly and expertise-intensive. MIMIR addresses this gap by **automating the hardware generation process, enabling non-expert users** to obtain efficient FPGA implementations directly from software-level specifications.

High-Level Synthesis (HLS)

Traditional hardware design relies on Hardware Description Languages (HDLs), which expose low-level structural and timing details of digital circuits. High-Level Synthesis (HLS) compilers raise the abstraction level by **translating C/C++-like programs into hardware**, guided by optimization directives (pragmas) similar to those used in OpenMP or CUDA. Despite this abstraction, effective HLS usage still requires significant expertise in memory hierarchies, parallelism, and hardware constraints. Recent systems, such as **SODA**, leverage **MLIR pipelines to translate machine learning models** (e.g., PyTorch, TensorFlow, ONNX) **into optimized hardware implementations**.

AI Agents

Recent advances in generative AI enable **automated code generation** with strong **reasoning** and **tool-use capabilities**. Modern agents can iteratively refine programs, interact with compilers, and integrate external tools, enabling closed-loop optimization workflows.



MIMIR: AI Agents for Hardware Generation

MIMIR combines a **multi-agent generative AI** system with **HLS toolchains** to **automate FPGA accelerator design**.

Given a reference CPU implementation, the system generates **functionally equivalent hardware designs, evaluates them**, and **iteratively explores the design space** to optimize QoR metrics such as latency, throughput, and resource utilization.

First, the agents analyze the codebase, understanding its key components and compilation and run commands. Second, the agents perform a runtime profile of the application, identifying key bottlenecks in the execution. Then the code is either optimized or translated into Torch, following a plan-implement-verify procedure.

In case of **translation into a Torch model**, the code can be easily ported across architectures. Leveraging **SODA** [1], MIMIR can **translate the Torch model into Verilog** by first translating it into TOSA MLIR, performing key optimizations, lowering it into LLVM IR, and finally synthesizing hardware using the Bambu HLS [2] LLVM entrypoint.

The optimization pipeline, instead, is integrated with the **A3D** [3] agentic workflow, which **translates the code from C into System C** and then synthetizes it into hardware using **Siemens Catapult**.

References

- [1] Nicolas Bohm Agostini, Serena Curzel, David Kaeli, and Antonino Tumeo. Soda-opt an mlir based flow for co-design and high-level synthesis. In Proceedings of the 9th ACM International Conference on Computing Frontiers, pages 201–202, 2022.
- [2] Christian Pilato and Fabrizio Ferrandi. Bambu: A modular framework for the high level synthesis of memory-intensive applications. In 2013 23rd International conference on field programmable logic and applications, pages 1–4. IEEE, 2013.
- [3] Abinand Nathallanbi, Anand Raghunathan. A3D. Ongoing research at Purdue University.

